

Design and Analysis of Algorithms 2025/26 – Project 2

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Reports should focus on the following points, clearly and succinctly, assuming the problem statement is known. In particular, you do not need to specify what each of the letters used in Section **Constraints** represents.

Page Limit: 5.

NOTE: All text in blue should be removed.

1 Problem Resolution

To start, in terms of data structures for our solution, we decided to use a two-dimensional array to store the positions of the beams. Moreover, we decided to use a graph implementation with adjacent lists (successors and predecessors). In terms of algorithms, we decided to implement a BFS search to filter any unnecessary nodes to the solution. Furthermore, Kahn's algorithm (topological sort) with built in acyclicity checking is used, so that disasters and beam removal order could be determined.

Firstly, we decided to incorporate a two dimensional array before beginning any operations with graphs or nodes. This was done so that beam dependancies could more easily and intuitively be made. The map is filled by iteration through every beam and filling the map based on the direction and length of the beam, filling up all coordinates that the beam belongs to with its integer ID.

0	0	6	0	0	0	0	0
0	5	6	0	4	0	0	0
0	5	1	1	4	0	0	0
0	5	0	3	2	2	2	2
0	0	0	3	0	0	0	0

Fig. 1. Filled in two dimensional array using problem description example

After the map has been filled, we iterate through each beam, starting from where it ends and checking the tiles in that direction until the end of the map. For every unique number it passes a new edge is added to the graph corresponding to the dependancy it has on other beams. For the graph implementation itself, we decided to use an adjacent list both with sucessor and predessesor data structures as both inAdjacentNodes and outAdjacentNodes need to be performed on BFS and Topological Sorting. This decision makes both these operations more efficient temporaly.

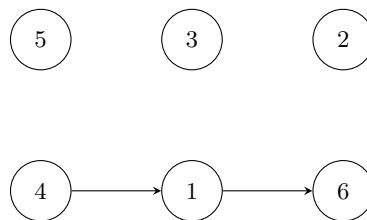


Fig. 2. Dependency graph of the problem description example

Subsequently, now that we have the graph to work with, we can process the graph using Breadth First Search and Topological Sort to get our final answer. We use BFS to find the nodes that necessarily need to be processed (beams to remove) given the columns you want to clear. This is done by executing `bfsExplore()` on any unfound nodes that belong to the columns you want to clear.

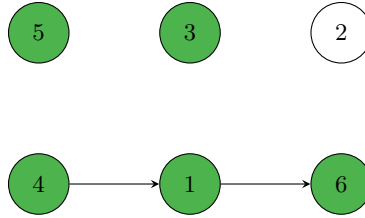


Fig. 3. Nodes processed by BFS (explored nodes are in green).

Once BFS is executed, we have a boolean array holding information about the nodes found, with the index of the array associating to if the node of that id was processed by BFS. An int array is also generated that holds the ids of the nodes that were found, this is used to iterate through the Topological Sort algorithm.

After BFS is executed, the Topological Sort algorithm is executed to remove the beams in the right order. Since the problem states that whenever multiple beams can be freed, the one with the lowest identifier must be freed, a priority queue was used.

The final solution for the problem description example would then be:

3 4 1 5 6

2 Time Complexity

Calculate the time complexity of the algorithm, justifying the parts and the result.

Use different “letters” to represent different “quantities”. If you do not use the letters given in the problem statement, do not forget to indicate what each letter represents.

3 Space Complexity

Calculate the space complexity of the program, justifying the parts and the result.

Use different “letters” to represent different “quantities”. If you do not use the letters given in the problem statement, do not forget to indicate what each letter represents.

4 Conclusions

List the strengths and weaknesses of your solution, the alternatives studied or that deserved to be studied, possible improvements, etc.

Clearly specify which websites were consulted and which AI tools were used, for what purposes, and how the obtained information was used, both to develop the program and to produce this report.